Where and Why Corn Fails in the Corn Belt

17 Years Climate Driven Crop Insurance Losses (2008-2024)

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INTRODUCTION

Drought and Excess Moisture are the leading cause of corn yield loss in the U.S. However, we still lack a clear picture of where production is chronically fragile to climate, why and whether that fragility is predictable.

The Risk Management Agency – Cause of Loss program provides how producers absorbs those losses. We treat the county insurance loss record as a revealed signal of climate fragility.

We read **17 years** of claims across **13 Corn Belt** states and ask where **corn** fails, why, and whether risk is predictable from loss history alone.

DATA

Cause of Loss (COL) Data

Risk Management Agency (RMA) publishes crop insurance data every year since 1989. This data covers entire U.S. There are 148 crop types and 31 variables.

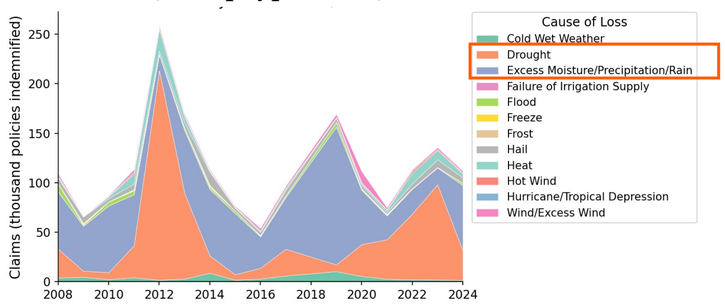
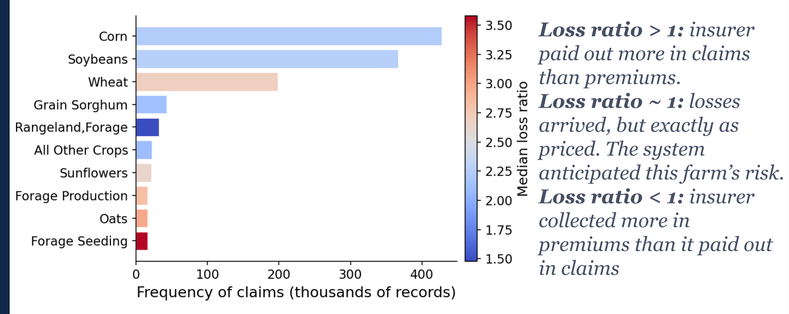


Fig.1: This stacked area plot shows the insurance claims for corn for different years (13 Corn Belt States). Excess moisture and Drought has the highest area in the plot.

Insurance data includes **Loss Ratio** column calculated as **indemnities / total premiums**



Loss ratio > 1: *insurer paid out more in claims than premiums.*
Loss ratio ~ 1: *losses arrived, but exactly as priced. The system anticipated this farm's risk.*
Loss ratio < 1: *insurer collected more in premiums than it paid out in claims*

Fig.2: Top 10 crops by claim frequency, colored by median loss ratio

Per insured acre, premiums and payouts barely change with farm size. Only liability moves. It peaks in mid-size counties (~$695) and collapses in the largest-farm counties ($387).

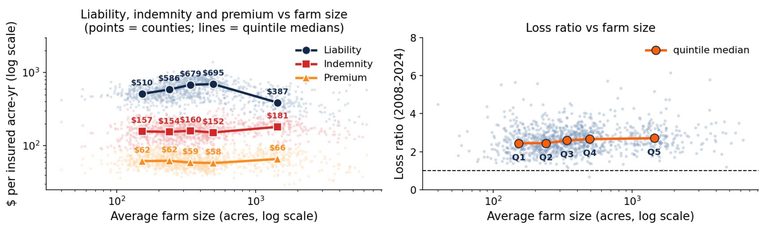


Fig.3: Farm size and corn insurance outputs across counties

SPEI Data

The Standardized Precipitation–Evapotranspiration Index is a multi-scalar drought index based on the climatic water balance. It captures both the supply and the atmospheric demand for water. Negative values indicate drier-than-normal conditions and positive values wetter-than-normal, with magnitudes beyond ±1.5 marking severe anomalies. SPEI6 data for September was downloaded for this study, to have the data for growing season. (*https://spei.csic.es*)

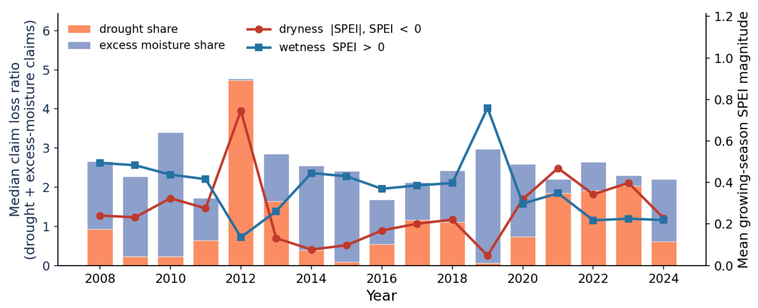


Fig.4: Median corn loss ratio by cause vs growing season dryness and wetness

Other Data Sourses and Final Panel

16,593 county-years, **1,080** counties, **13** Corn Belt states,

- RMA Cause-of-Loss (https://www.rma.usda.gov/tools-reports/summary-of-business/cause-loss),
- gridMET (https://www.climatologylab.org/gridmet.html)/SPEI,
- SoilGrids (https://soilgrids.org),
- Census of Agriculture (https://www.nass.usda.gov/AgCensus/).
- References are listed in the QR code

Loss Ratio Landscape

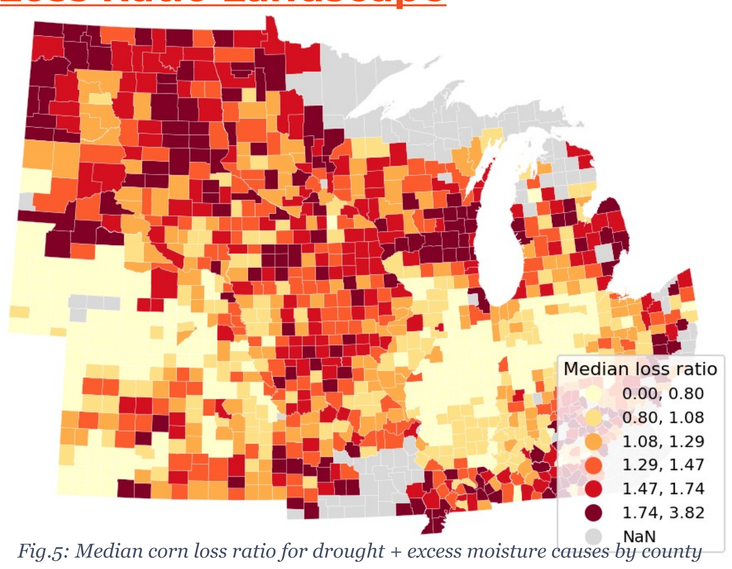


Fig.5: Median corn loss ratio for drought + excess moisture causes by county

METHODS

County Year Loss Ratio Matrix X
16,593 county-years + sliamte/soil/farm covariates

Branch A: Explanatory

A1: ESDA
Moran's I, LISA, Getis-Ord Gi
Clustering
Hot Spot Persistence

A2: Spatial regression
SAR / SEm (LM test)
GWR
Spatial non-stationarity

A3: Panel FE Decomposition
Structural (17%)
Episodic (83%)
RF predictability of each

Branch B: Similarity and Prediction

B1: County Similarity Kernel
Gram Matrix K
Dot product of loss histories
$K_{i,j} = \sum X_{i,t}\ X_{j,t}$

Do LISA Clusters = embedding clusters?

B2: Spatial Embedding
Counties → (dim1, dim2)
Risk Profile Map in 2D

B3: Climate Interpretation
Dim1 vs temp, precip, cool/dry axis

Compare R2, RF and kernels

B4: Kernel Prediction
SVR + GPR
Point Forecast
Uncertainty Bands

SYNTHESIS: Where risk sits (A1), Why it varies (A2, A3)
Which counties behave alike and what to expect next (B1-B4)

RESULTS | Branch A

Exploratory Spatial Data Analysis, A1

Losses cluster in every year(Moran's I = 0.33–0.71, p= 0.001), peaking in the 2012 drought and wet 2019.
Hot spots (246 counties, red) are whose neighborhoods run high loss ratio. These are structural-risk story in significance-tested form.
Cold spots (245 counties, blue) are two big islands of statistically low risk. The Corn Belt's core is a cold spot of climate losses.

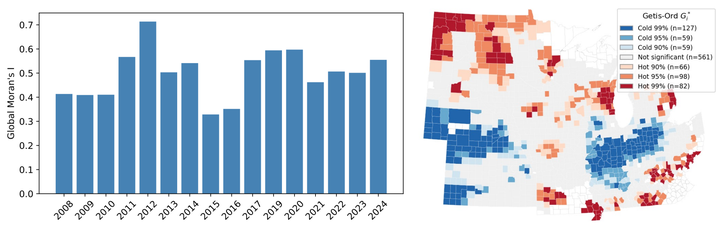
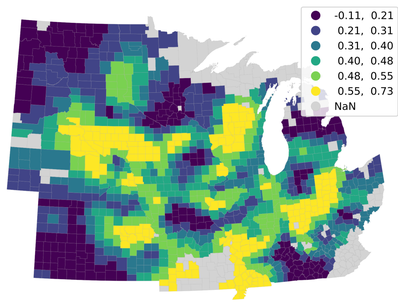


Fig.6: (left) Moran's I by year, (right) hot and cold spots of median corn loss ratio 2008-2024

Spatial Regression, A2



Global OLS fits the loss surface poorly (R^2 = 0.25 in 2012) and leaves strongly autocorrelated residuals. GWR raises fit (0.76 in 2012)

Fig.7: Local GWR, 2012

Panel Fixed Effect, A3

Only **17%** of loss-ratio variance is between-county (structural); **83%** is within-county, driven by the year's weather. Both layers carry signal. Random forests predict the structural county mean at cross-validated R^2 = 0.47 and annual anomalies at R^2 = 0.37.

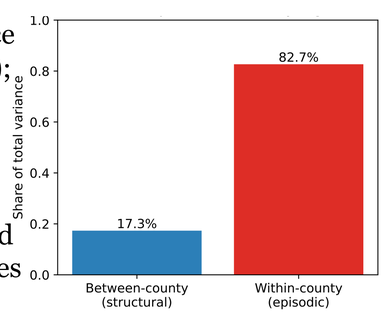


Fig.8: Variance Decomposition of county log loss ratio

RESULTS | Branch B

Gram-matrix encodes pairwise similarity between every two counties based on their year-by-loss-ratio profiles. This can uncover drivers of why certain counties share risk profiles. High value means they lost in the same years with similar severity.

Spatial Embedding B2

Spatial embedding compresses loss histories into two data-driven axes.

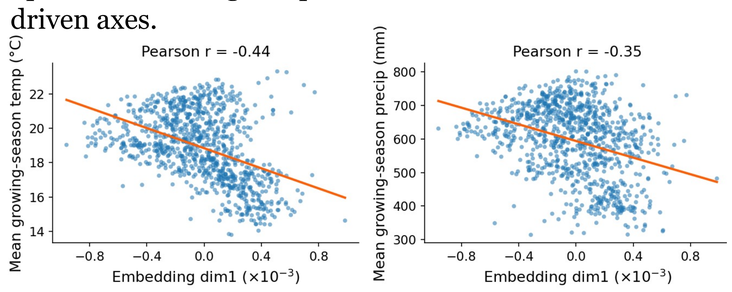


Fig.9: County similarity embeddings tracks climate.

Climate Interpretation, B3

The model never saw any weather data, only where and when farmers lost money. Yet the main axis of its county map turns out to follow temperature and rainfall ($r = -0.44$ / -0.35). Insurance claims alone help to map the Corn Belt's climate fragility map.

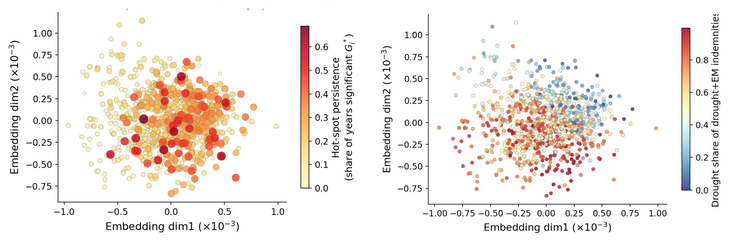


Fig.10: County similarity embeddings, colored by (left) hot spots, (right) drought driven – excess moisture driven counties

Kernel Prediction , B4 | Support Vector Regression & Gaussian Process Regression

Within the observed period, loss histories pin down county risk almost exactly (SVR RMSE = **0.160**, GPR RMSE = **0.152)**. Looking forward, they still rank counties well (r = 0.65, RMSE ~0.53 vs. 0.66 baseline), but most of the remaining error is next year's weather, which is unpredictable in principle. The history tells where the fragile counties are.

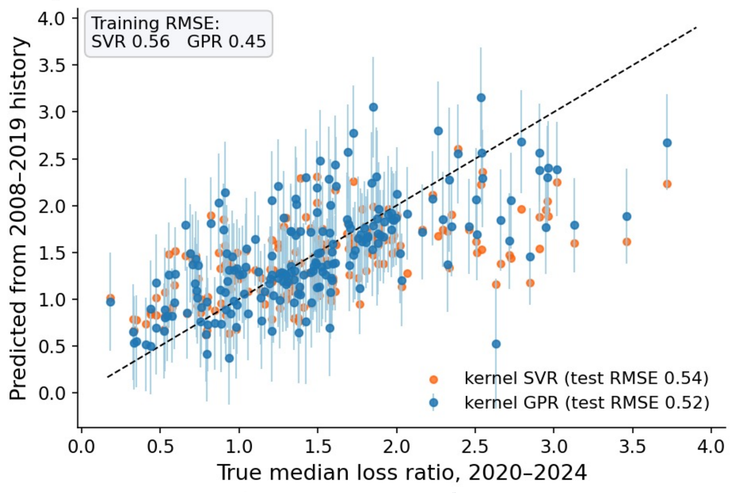


Fig.11: Forecast test results

CONCLUSION

This study turned 17 years of insurance claims into a map of Corn Belt climate fragility: *where corn fails, why it fails there, who bears the cost, and how far ahead we can see it* coming. The chronic counties are a ready target list for drought tolerance in the west and drainage in the wet corridors, and spatially varying premium rating would make the program fairer. We cannot time the disaster, but we can know which counties it will find.

Acknowledgments

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